Instacart E-Commerce Analytics Final Report

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Tools: Databricks (SQL), Microsoft Excel
Data Source: Kaggle – Instacart Online Grocery Shopping Dataset
Project Repository: https://github.com/Princemoto/instacartportfolioproject
Submitted to: The Brickshub Analytics Team Lead
Date: March 2026

1. Executive Summary

Customers demonstrate strong, repeated shopping habits with predictable weekly trends. Orders peak on Sundays and Mondays between 10 AM and 3 PM. Fresh produce drives most demand, led by bananas. Most customers reorder within 4–7 days, reflecting consistent weekly grocery shopping patterns. These insights support better inventory planning, marketing strategies, and customer engagement.

2. Project Overview

This project analyses customer purchasing behaviour in the Instacart online grocery platform using large-scale transactional data. The objective is to identify ordering patterns, customer segments, and product demand trends that can inform operational and marketing decisions. Analysis was conducted using Databricks SQL, with results validated in Excel and visualised through Power BI dashboards.

3. KPI Summary

KPI	Result
Total Orders	**3.21M**
Total Customers	**206.21K**
Avg Orders per Customer	**15.59**
Reorder Rate	**58.97%**
Peak Order Days	**Sunday, Monday**
Peak Order Hours	**10 AM – 3 PM**
Avg Days Between Orders	**10.7 days**
Most Common Reorder Cycle	**4–7 days**
Top Product	**Banana**
Top Department	**Produce**

Note: Metrics validated with a 5,000-customer Excel sample

4. Customer Behavioural Insights

When people order: Most orders come in between 10 AM and 3 PM, with Sundays and Mondays the busiest. Midweek orders are noticeably lower.

Repeat behaviour: Using the days_since_prior_order column, orders were segmented as First order (Null values), 0–3 days, 4–7 days, 8–14 days, 15–21 days, and 22–30 days. Most customers reorder within 4–7 days, showing a clear weekly pattern.

Who's shopping: Customers were grouped by total number of orders: first-time (1–2), new (3–5), growing (6–10), regular (11–20), and power users (20+). The largest group is new (3–5), indicating that most customers are still early in their purchasing journey.

The charts below show the Instacart customers' behavioural patterns:

Chart 1: Orders by Day

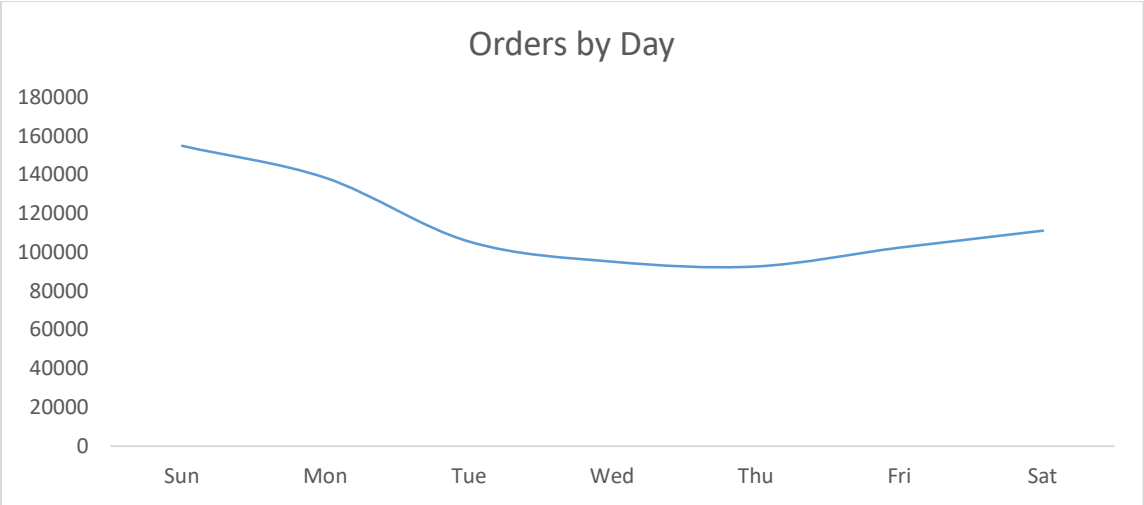


Chart 2: Orders by Hour of Day

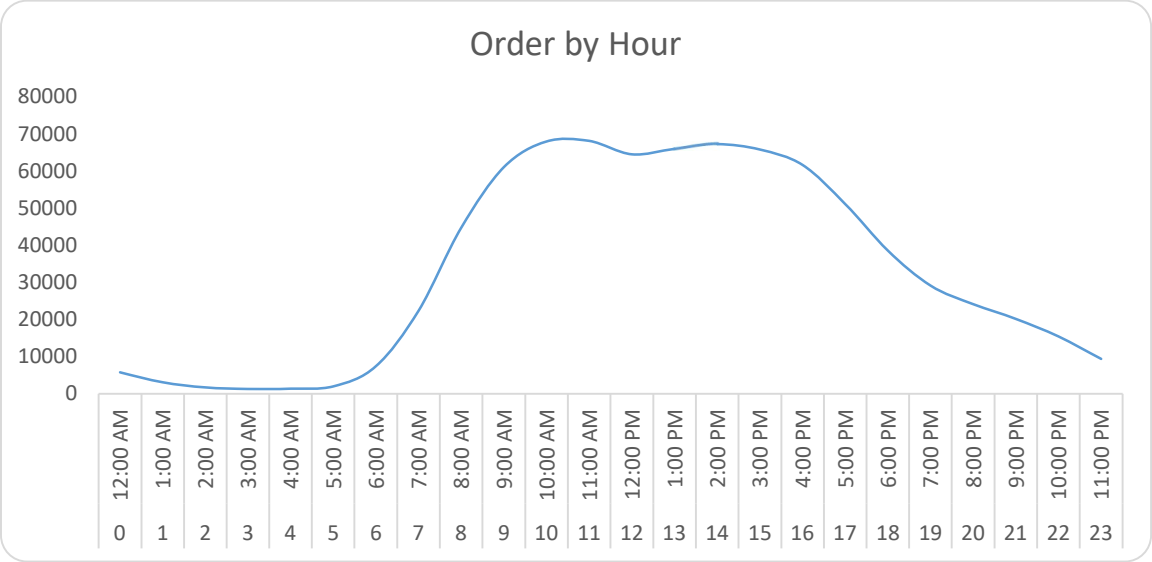
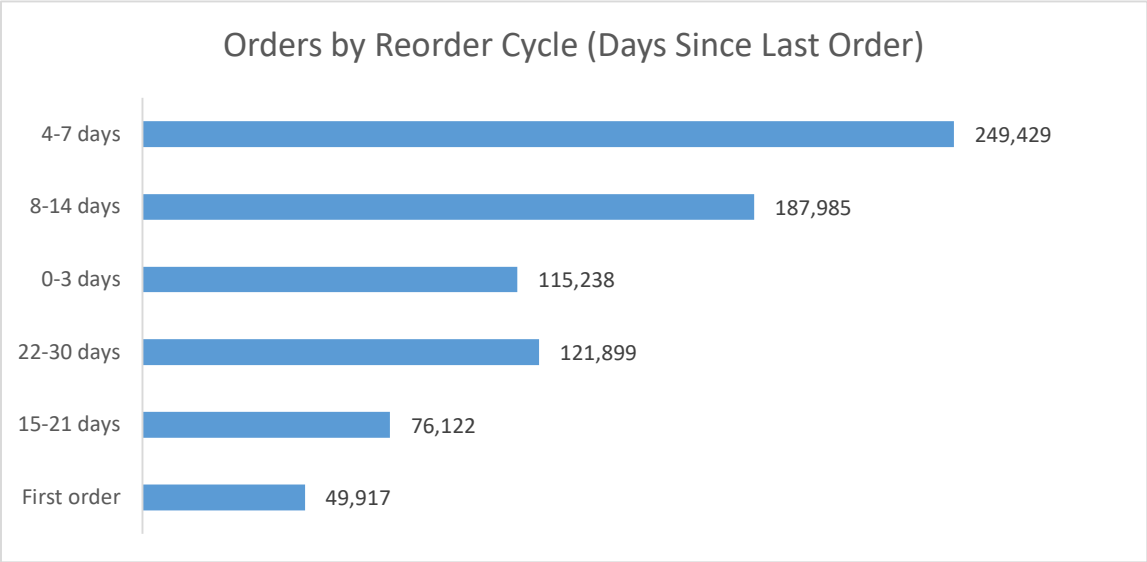


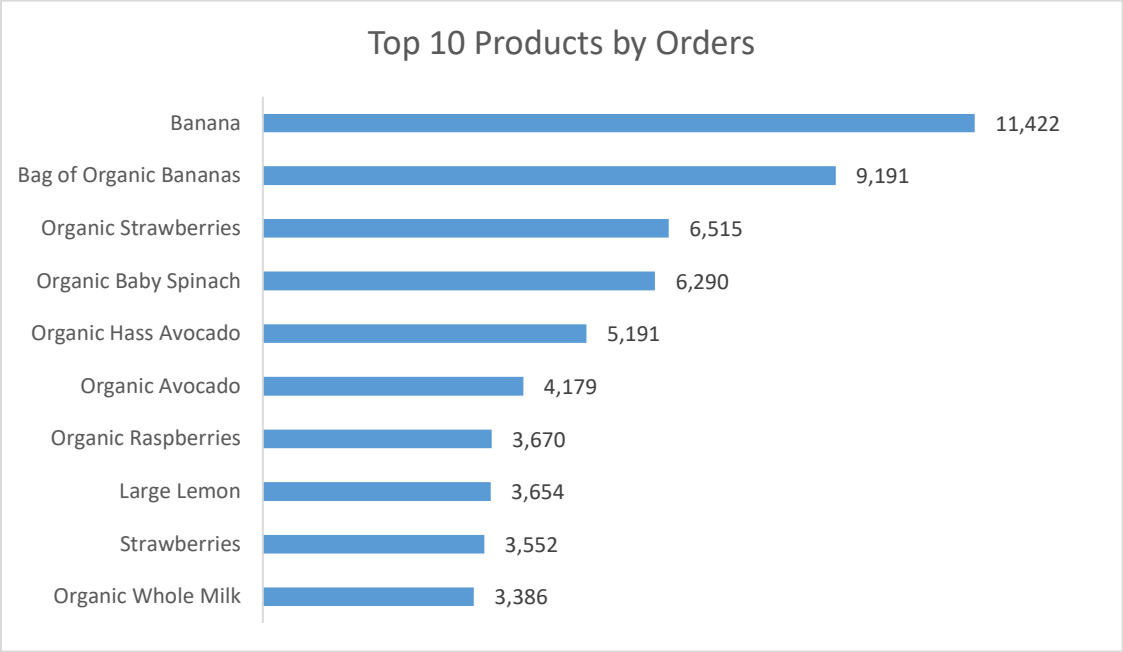
Chart 3: Reorder Cycle Distribution



5. Product and Department Insights

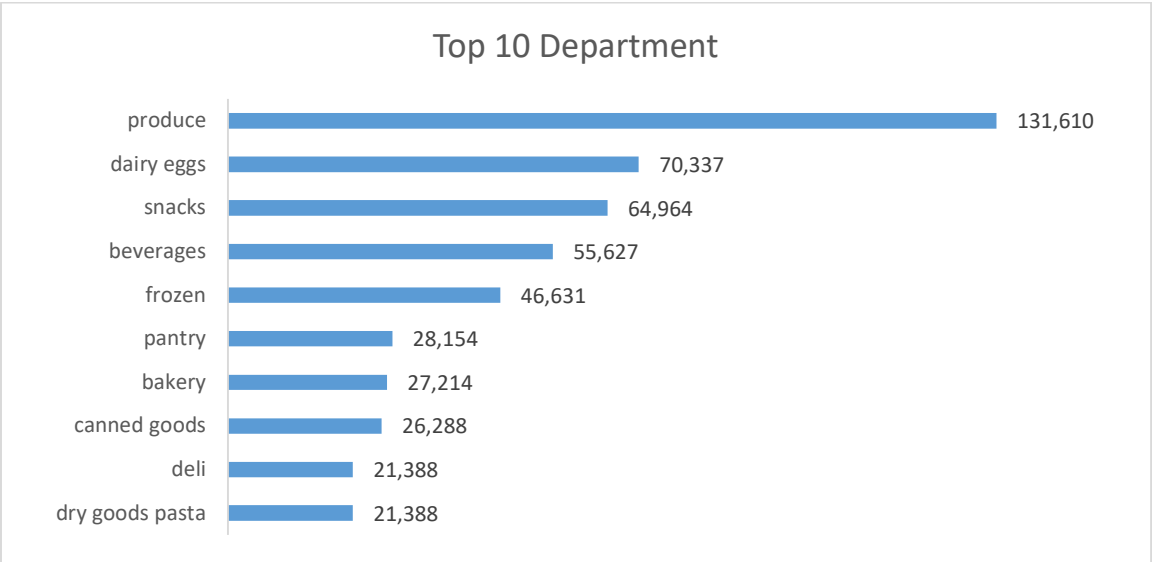
Bananas lead overall product orders, highlighting strong demand for staple grocery items. See the chart below:

Chart 5: Top 10 products by Orders



The Produce department has the highest order volume, indicating that fruits and vegetables constitute a significant portion of purchases.

Chart 6: Top 10 Departments by Orders



6. Technical Architecture

The project follows the **Databricks Medallion Architecture**, which structures the data into three layers for better data quality and usability.

- **Bronze Layer:** Raw Instacart data was ingested and stored without modification to preserve the original dataset.
- **Silver Layer:** Data cleaning and transformations were applied, including joins across orders, products, departments, and customer data to create structured analytical tables.
- **Gold Layer:** Aggregated tables and key metrics were generated using Databricks SQL to support business analysis.

The curated datasets were then connected to **Power BI**, where an interactive dashboard was developed to visually explore customer behaviour, ordering patterns, product demand, and key performance indicators. See the appendix for Power BI screenshots. SQL queries and supporting files for this project are available in the GitHub repository.

7. Recommendations

Based on the analysis, several strategic opportunities emerge.

Inventory Management
Focus on keeping produce and fresh goods well-stocked, especially on weekends and Mondays when orders peak.

Marketing Strategy
Promotions work best during the busiest hours (10 AM to 3 PM) and early in the week when demand is highest.

Customer Engagement
Loyalty incentives can encourage repeat purchases, and showcasing frequently purchased products keeps customers engaged.

Product Placement
Make sure high-demand staples like bananas and other fresh produce are prominent, and pair them with complementary items to encourage additional purchases.

8. Conclusions

The analysis shows that Instacart customers follow predictable grocery shopping patterns. Many customers reorder within a weekly cycle, and fresh produce, particularly bananas,

accounts for a large share of orders. Ordering activity also peaks between late morning and early afternoon, especially on Sundays and Mondays.

Customer segmentation and reorder cycle analysis further highlight how different groups interact with the platform, from first-time buyers to frequent shoppers.

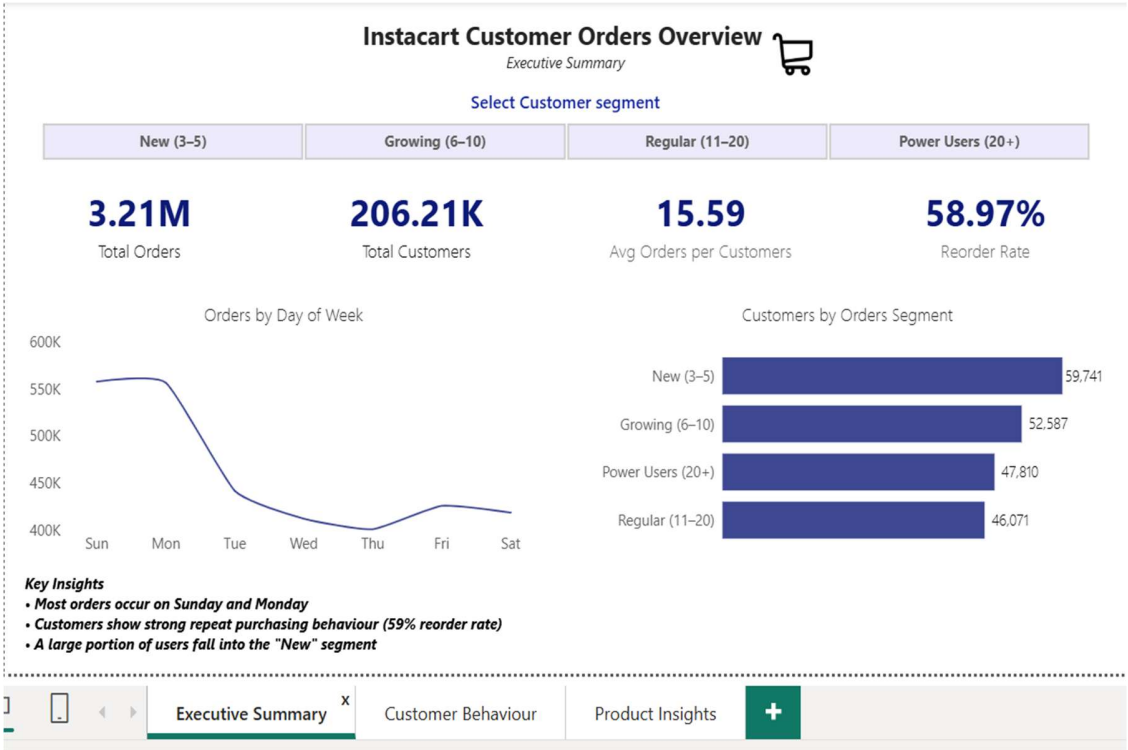
From a technical standpoint, the project used the Databricks Medallion Architecture to structure the data pipeline, with analysis performed in Databricks SQL and insights visualised through a Power BI dashboard.

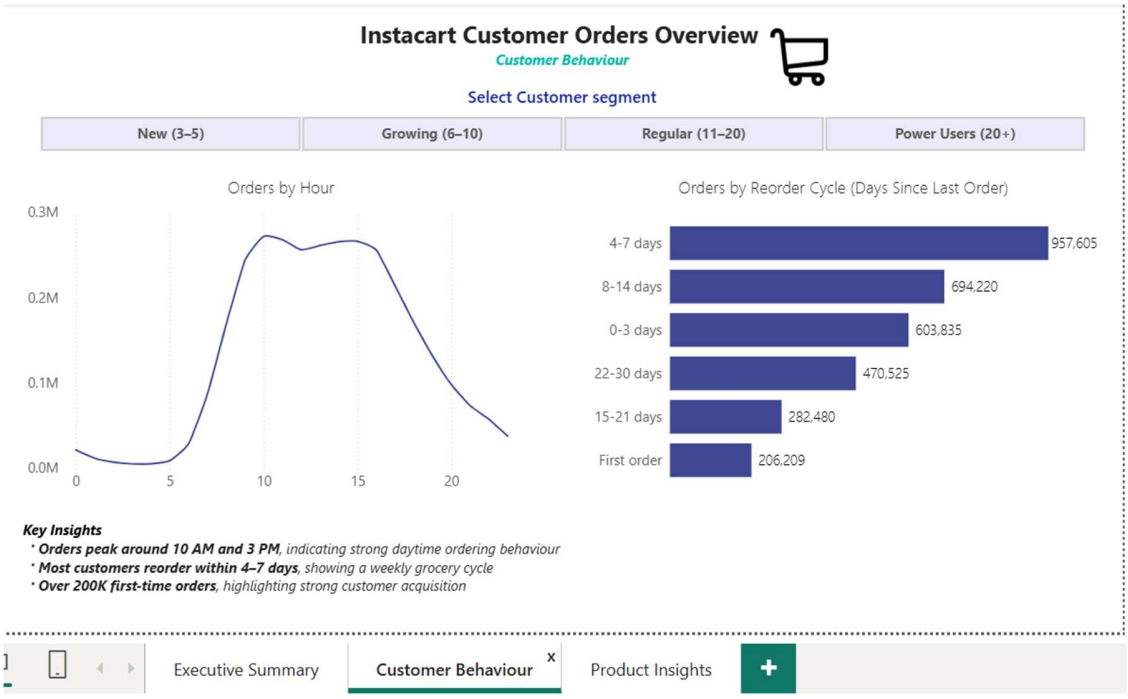
Overall, the project demonstrates how combining data engineering and analytics can reveal practical customer behaviour insights and support more informed operational decisions.

Appendix: Power BI dashboard

The following screenshots show the interactive Power BI dashboard developed to visualise customer behaviour, ordering patterns, and key performance indicators.

Screenshot 1: KPI Overview



Screenshot 2: Customer Behaviour Dashboard

Screenshot 3: Product Demand Dashboard